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RHEA SUKTHANKER

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RESEARCH INTERESTS

My research focuses on automating and optimizing foundation model inference—particularly for Large Language Models (LLMs) and vision models—to enable efficient, scalable, and cost-effective deployment in real-world applications. While training these models incurs significant one-time compute, their inference costs dominate long-term energy consumption, latency, and memory usage across deployment contexts. To address this challenge, my work develops novel and automated techniques for model compression, including structured and unstructured pruning, low-bit quantization, and knowledge distillation, with an emphasis on reducing manual hyperparameter tuning and expert intervention. By integrating hardware-aware optimization, automated algorithm selection, and scalable search strategies, I aim to make foundation models more accessible, sustainable, and practical for diverse domains ranging from edge computing to large-scale cloud services.

As part of this goal, I actively contribute to open-source tools and research infrastructure. I co-developed the library [whittle](#), a framework for automated model compression of large language models, and contribute to the neural architecture search library [NASLib](#), supporting research on reproducible architecture search methods.

My recent research interests include:

- [Automated Foundation Model Compression](#): Developing scalable, hardware-aware algorithms for pruning, low-bit quantization and knowledge distillation.
- [Efficient and Multi-Objective Neural Architecture Search](#): Designing gradient-based, weight-sharing, and multi-objective optimization techniques for scalable neural architecture search under compute and memory constraints.

EDUCATION

Department of Computer Science, University of Freiburg

Freiburg, Germany

Ph.D. in Computer Science

Feb 2022 - May 2026

- Advisor: [Prof. Frank Flutter](#)
- Research focus: Efficient and Scalable Multi-Objective Optimization for Foundation Models

Department of Computer Science, ETH Zurich

Zurich, Switzerland

Masters in Data Science

Sept 2018 - July 2021

- GPA: 5.39/6
- Awarded the [ESOP](#) Scholarship.

Department of Information Technology, VIT University

Vellore, India

Bachelor's in Information Technology

June 2014 - June 2018

- GPA: 9.75/10; Rank: 2nd
- Awarded merit scholarship for academic excellence.

PROFESSIONAL EXPERIENCE

Microsoft

Redmond, WA, USA

Senior Researcher, Applied Sciences Group

June 2026 - Present

- Research on [Microsoft Aion](#), a family of on-device AI models for Windows 11.
- Completed Ph.D. at the University of Freiburg, advised by [Prof. Frank Hutter](#), in 2026.

INTERNSHIP EXPERIENCE

Microsoft Research

Cambridge, UK

Research Intern (Applied Sciences Group)

May 2025 - July 2025

- Supervisors: [Dr. Pashmina Cameron](#) and [Dr. James Hensman](#)

- Project: **Automated Quantization of LLMs (AQUA)**
- Research Contributions:
 - Contributed to AQUA, an internal Microsoft Research framework for automated end-to-end post-training quantization of large language models (LLMs); designed and implemented a LoRA + knowledge distillation fine-tuning pipeline enabling stable, parameter-efficient adaptation of highly compressed (2-bit) models without reverting to higher-precision weights.
 - Advanced Microsoft’s internal 2-bit vector quantization framework by integrating LoRA A/B matrices directly into the quantized setting, investigating dataset mixing strategies for next-generation Phi models, and conducting extensive empirical studies on fine-tuning and distillation to derive best practices for training ultra-compressed LLMs under extreme memory and hardware constraints.
 - Demonstrated performance surpassing Quantization-Aware Training (QAT) methods at a fraction of the computational and training cost, significantly improving the practicality of 2-bit model deployment; contributing inventor on the patent “Vector Quantization using a Learnable Codebook.”

RESEARCH EXPERIENCE

Computer Vision Lab, ETH Zurich

Zurich, Switzerland

Student Researcher

March 2021 - April 2022

- Advisors: [Dr. Zhiwu Huang](#) and [Dr. Suryansh Kumar](#)

- Research Contributions:
 - Neural Architecture Search of SPD Manifold Networks (Master’s Project , ETH Zurich)**
 - Formulated a new neural architecture search (NAS) problem for Symmetric Positive Definite (SPD) manifold networks and defined a geometry-aware search space tailored to non-Euclidean data representations.
 - Developed a one-shot NAS method using a differentiable supernet to efficiently explore SPD architecture candidates while preserving manifold structure.
 - Demonstrated improved performance over state-of-the-art handcrafted SPD networks and traditional NAS algorithms on multiple vision benchmarks with significantly lighter architectures.

Generative Flows with Invertible Attentions (Master’s Thesis, ETH Zurich)

- Proposed novel normalizing flow architectures with invertible attention mechanisms to improve expressivity while preserving exact likelihood computation.
- Conducted large-scale experiments showing improved generative modeling performance over baseline flow models.

- Resulted in publication at *Computer Vision and Pattern Recognition (CVPR) 2022*.

Computational Intelligence Laboratory, NTU

Singapore

Research assistant

May 2017-July 2017 and Jan 2018 – May 2018

- Advisor: [Dr. Erik Cambria](#)
- Research Contributions:
Anaphora and Coreference Resolution: A Review
 - Authored a comprehensive survey of coreference and anaphora resolution methods, analyzing rule-based, feature-based, and neural approaches.
 - Synthesized evaluation metrics, benchmark datasets, and emerging research trends to highlight challenges and future directions in document-level NLP.

PREPRINTS

1. Arjun Krishnakumar*, [Rhea Sanjay Sukthanker*](#), Hannan Javed Mahadik*, Gabriela Kadlecová, Vladyslav Moroshan, Timur Carstensen, Frank Hutter, Aaron Klein. [Where to Begin: Efficient Pretraining via Subnetwork Selection and Distillation](#). (under review).

JOURNAL PUBLICATIONS

1. [Rhea Sukthanker](#), Soujanya Poria, Erik Cambria, Ramkumar Thirunavukarasu. [Anaphora and coreference resolution: A review](#). *Information Fusion (IF:15.5)*.

WORKSHOP PUBLICATIONS

1. [Rhea Sukthanker](#), Benedikt Staffler, Frank Hutter, Aaron Klein. [Large Language Model Compression with Neural Architecture Search](#). *NeurIPS 2024 Compression Workshop*.
2. [Rhea Sukthanker*](#), Arber Zela*, Benedikt Staffler, Samuel Dooley, Josif Grabocka, Frank Hutter. [Multi-Objective Differentiable Architecture Search](#). *ICML 2024 WANT Workshop*.
3. Yan Wu, Zhiwu Huang, Suryansh Kumar, [Rhea Sanjay Sukthanker](#), Radu Timofte, Luc Van Gool. [Trilevel Neural Architecture Search for Efficient Single Image Super-Resolution](#). *CVPR 2022 NAS Workshop*.

CONFERENCE PUBLICATIONS

*: equal contribution

1. Arjun Krishnakumar*, [Rhea Sanjay Sukthanker*](#), Hannan Javed Mahadik*, Gabriela Kadlecová, Vladyslav Moroshan, Timur Carstensen, Frank Hutter, Aaron Klein. [whittle: A Library for Sub-Network Extraction from Large Language Models](#). *International Conference on Automated Machine Learning (AutoML 2026)*, New York, USA (Best Paper Nominee, Oral).
2. [Rhea Sukthanker*](#), Arber Zela*, Benedikt Staffler, Samuel Dooley, Josif Grabocka, Frank Hutter. [Multi-Objective Differentiable Architecture Search](#). *International Conference on Learning Representations (ICLR 2025)*, Singapore.
3. [Rhea Sukthanker](#), Arber Zela, Benedikt Staffler, Aaron Klein, Lennart Purucker, Jörg K. H. Franke, Frank Hutter. [HW-GPT-Bench: Hardware-Aware Architecture Benchmark for Language Models](#). *Neural Information Processing Systems DBT Track (NeurIPS 2024)*, Vancouver, Canada.

4. **Rhea Sukthanker**, Arjun Krishnakumar, Mahmoud Safari, Frank Hutter. [Weight-Entanglement Meets Gradient-Based Neural Architecture Search](#). *International Conference on Automated Machine Learning (AutoML 2024)*, Paris, France.
5. Samuel Dooley*, **Rhea Sukthanker***, John P. Dickerson, Colin White, Frank Hutter, Micah Goldblum. [Rethinking bias mitigation: Fairer architectures make for fairer face recognition](#) oral. *Neural Information Processing Systems (NeurIPS 2023)*, New Orleans, USA.
6. Simon Schrodli, Danny Stoll, Binxin Ru, **Rhea Sukthanker**, Thomas Brox, Frank Hutter. [Construction of Hierarchical Neural Architecture Search Spaces based on Context-free Grammar](#). *Neural Information Processing Systems (NeurIPS 2023)*, New Orleans, USA.
7. **Rhea Sukthanker**, Zhiwu Huang, Suryansh Kumar, Radu Timofte, Luc Van Gool. [Generative flows with invertible attentions](#). *Computer Vision and Pattern Recognition (CVPR 2022)*, New Orleans, USA.
8. **Rhea Sukthanker**, Zhiwu Huang, Suryansh Kumar, Radu Timofte, Luc Van Gool. [Neural Architecture Search of SPD Manifold Networks](#). *International Joint Conferences on Artificial Intelligence (IJCAI 2021)*, Montreal, Canada.

PATENTS

1. A. Zela, B. S. Staffler, F. Hutter, M. Safari, and **R. S. Sukthanker** (Feb. 19, 2025b). “Method and/or apparatus for architecture search”. U.S. pat. req. US20250272576A1. Robert Bosch GmbH. URL: <https://patents.google.com/patent/US20250272576A1/>. Published.
2. A. Zela, B. S. Staffler, F. Hutter, M. Rapp, and **R. S. Sukthanker** (May 19, 2025a). “Method and device for reducing a network dimension of a base model”. U.S. pat. req. US20250378332A1. Robert Bosch GmbH. URL: <https://patents.google.com/patent/US20250378332A1/>. Published.
3. R. Grazi, **R. S. Sukthanker**; E. Portugaly, H. R. Jackson-FLux, P. J. Cameron, and J. J. Hensman.(Feb 2, 2026). “Vector Quantization using a Learnable Codebook”. U.S. pat. (filed)

ACADEMIC SERVICES

- [NeurIPS](#): 2023, 2024
- [ICML](#): 2024, 2025, 2026
- Reviewer**
 - [ICLR](#): 2024, 2025, 2026
 - [AutoML](#): 2024

Diversity and Inclusion Chair

- [AutoML](#) 2024

Teaching

- Deep Learning Lab (Semester Course: 2022)
- Foundations of Deep Learning (Semester Course: 2023, 2024)
- Pruning and Efficiency in Large Language Models (Seminar Course: 2024)

AWARDS AND HONORS

- Awarded [Goa Scholars 2018-19](#)
- Awarded [ETH Zurich Excellence Scholarship](#)
- 1st place in [AutoML Cup](#) organized at AutoML 2023
- 3rd place in [AutoML Decathlon](#) organized at NeurIPS 2022
- [Oral Presentation](#) at NeurIPS 2023

INVITED TALKS

- [NeurIPS 2023 Oral Talk](#): "Rethinking bias mitigation: Fairer architectures make for fairer face recognition"
- [AutoML Seminar 2024](#) : "Rethinking bias mitigation: Fairer architectures make for fairer face recognition"

REFERENCES

[Prof. Dr. Frank Hutter](#): Prior Labs, ELLIS Institute Tübingen and University of Freiburg
Email: fh@cs.uni-freiburg.de

[Dr. Aaron Klein](#): ELLIS Institute Tübingen
Email: kleiaaro@gmail.com

[Dr. Zhiwu Huang](#): University of Southampton, UK
Email: Zhiwu.Huang@soton.ac.uk

[Dr. Suryansh Kumar](#): Texas A&M University College Station, USA
Email: suryanshkumar@tamu.edu

[Dr. Pashmina Cameron](#): Microsoft, Redmond, Seattle, USA
Email: pcameron@microsoft.com

[Dr. James Hensman](#): Microsoft Research, Cambridge, UK
Email: jameshensman@microsoft.com